

The Photon Non-Paradox Theorem: Resolution of Wave-Particle Duality, Self- Interference, Delayed Choice, and Measurement Collapse in Phase-Primitive Frameworks

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Abstract

We prove the Photon Non-Paradox Theorem (PNPT), which establishes that the standard "photon paradoxes" — wave-particle duality, single-photon self-interference, delayed-choice retrocausality, which-path complementarity, and measurement-induced collapse — do not arise as genuine contradictions within phase-primitive theoretical frameworks, specifically Phase Theory, Light-Phase Substrate Theory (LPST), and Luminous Substrate Theory (LST). The theorem demonstrates that these apparent paradoxes are representational artifacts produced by importing object-trajectory ontology into frameworks where the photon is not an object but an equivalence class of admissible phase excitations. We formalize six definitions grounding the phase-primitive treatment of light, state and prove the theorem through four lemmas addressing trajectory identity, which-path decoherence, boundary-termination detection, and delayed-choice boundary reclassification, and derive four corollaries resolving each classical paradox. The result provides a unified, paradox-free account of all standard quantum-optical phenomena within a single ontological framework, without invoking wavefunction collapse, observer-dependent reality, or retrocausal influences. Experimental implications and connections to decoherence theory are discussed.

1. Introduction and Motivation

The quantum theory of light has been, since its inception, the most conceptually fertile and most paradox-ridden domain of fundamental physics. When Albert Einstein proposed in 1905 that light consists of discrete energy quanta — *Lichtquanten* — he did not merely offer a solution to the photoelectric effect; he inaugurated a century of conceptual struggle over the nature of the photon [1]. Within two decades, Niels Bohr's complementarity principle (1928) crystallized the apparent contradiction at the heart of quantum optics: light behaves as a wave in some experimental contexts and as a particle in others, and no single classical description encompasses both aspects [2]. Paul Dirac, in his foundational 1930 treatise, elevated this tension to the status of fundamental principle, declaring that "each photon interferes only with itself" — a statement that has provoked confusion and debate for nearly a century [3]. Richard Feynman, in his celebrated 1965 Messenger Lectures, went further, famously characterizing the double-slit experiment as embodying "the only mystery" of quantum mechanics [4].

What is remarkable is not that these conceptual difficulties were identified early in the history of quantum theory, but that they have persisted, essentially unresolved at the foundational level, through every subsequent decade of extraordinary experimental and theoretical progress. The paradoxes of the photon are not artifacts of an immature theory; they are embedded in the mature formalism of quantum electrodynamics, quantum information theory, and quantum optics. No amount of calculational success has dissolved them. They are, in a precise sense, structural features of the representational framework, not empirical puzzles awaiting better data.

The present paper addresses five specific paradoxes, each of which has generated an extensive literature and none of which has received a universally accepted resolution within the standard interpretive frameworks of quantum mechanics:

(a) Wave-particle duality. Light exhibits mutually exclusive wave-like and particle-like behavior depending on the experimental arrangement. In diffraction and

interference experiments, light displays wavelength, phase, and superposition — canonical wave properties. In the photoelectric effect, Compton scattering, and photon-counting experiments, light arrives in discrete, indivisible quanta — canonical particle properties. The standard formalism encodes both behaviors but provides no unified ontological account of how a single entity can possess both characters. Bohr's complementarity principle [2] asserts that both descriptions are necessary and mutually exclusive, but this is widely regarded as a statement about the limits of classical concepts rather than a positive account of what light *is*.

(b) Single-photon self-interference. When the intensity of a light source is reduced to the single-photon level, interference patterns still build up one detection event at a time, as demonstrated conclusively by Taylor (1909) [33], Grangier, Roger, and Aspect (1986) [34], and numerous subsequent experiments. Dirac's dictum that each photon "interferes only with itself" [3] captures the empirical fact but raises the question: how does an indivisible quantum — a single particle — interfere with itself? If it is truly indivisible, it cannot be in two places at once. If it can be in two places at once, it is not indivisible in the ordinary sense. The formalism resolves this through the superposition principle, but the ontological puzzle remains.

(c) Wheeler's delayed-choice experiment. In 1978 and 1984, John Archibald Wheeler proposed a thought experiment — subsequently realized in the laboratory by Jacques *et al.* (2007) [11] and Manning *et al.* (2015) [38] — in which the experimenter's choice of measurement apparatus is made *after* the photon has "already" entered the interferometer [5, 6]. If the second beam splitter is inserted, interference is observed (wave behavior). If it is removed, which-path information is obtained (particle behavior). The paradox: the photon's behavior — wave or particle — appears to be determined retroactively by a future measurement choice. This has been interpreted as evidence for retrocausality, for the absence of a definite past, or for the fundamentally observer-dependent character of quantum reality. Wheeler himself drew the striking conclusion that "no phenomenon is a phenomenon until it is an observed phenomenon" [6].

(d) Which-path complementarity. In a two-path interferometer, if one introduces a device that records which path the photon takes, the interference pattern disappears. Removing the which-path information restores the pattern. This was

demonstrated with exquisite precision by Scully, Englert, and Walther (1991) [8] and formalized quantitatively by Englert's (1996) fringe-visibility/which-way inequality $D^2 + V^2 \leq 1$ [9], building on the earlier work of Greenberger and Yasin (1988) [16]. The paradoxical flavor arises from the impression that mere "observation" or "knowledge" of the path is what destroys the interference — as though the photon responds to what the experimenter knows, not to what physical interactions have occurred.

(e) Measurement-induced collapse. In the standard Copenhagen interpretation, detection of a photon causes an instantaneous, non-unitary transition from a superposition state to a definite outcome — the "collapse of the wavefunction." This postulate, introduced by von Neumann in 1932 [20] and formalized in the projection postulate, has no analogue in any other physical theory. It introduces an asymmetry between "measurement" and "ordinary" interactions that has never been satisfactorily defined, leading to the measurement problem — the central conceptual difficulty of quantum mechanics. The measurement problem has motivated all major alternative interpretations: Everett's many-worlds interpretation (1957) [22], Bohm's pilot-wave theory (1952) [21], decoherence-based approaches (Zeh 1970 [25]; Zurek 2003 [12]; Joos *et al.* 2003 [13]), QBism (Fuchs 2010) [23], and relational quantum mechanics (Rovelli 1996) [24].

The central claim of this paper is that these five paradoxes share a common structural origin. They do not arise from deep features of nature. They arise from a specific ontological commitment that is imported, often tacitly, into the description of photon behavior: the commitment to *object-trajectory ontology*. In object-trajectory ontology, a photon is a localized, persistent thing — an object — that travels a definite path through space and arrives at a definite location. Wave-like behavior is then anomalous: how can a thing be in two places? How can a thing interfere with itself? How can a future choice determine a past trajectory? How can observation change a thing's behavior? How can detection instantaneously alter a thing's state?

Every one of these questions presupposes that the photon *is* a thing with a trajectory. Remove that presupposition, and the questions dissolve — not because they are answered, but because they are no longer well-formed.

This paper demonstrates this dissolution rigorously within three recently proposed phase-primitive theoretical frameworks: *Phase Theory* [30], *Light-Phase Substrate Theory* (LPST) [31], and *Luminous Substrate Theory* (LST) [32]. These three theories, developed by the present author, differ in their detailed mathematical structures but share a common ontological foundation: the fundamental entity is not an object but a *phase-bearing substrate field*. In these frameworks:

- The photon is not an object; it is an *equivalence class of admissible phase configurations*.
- Interference is not self-action; it is *global admissibility* — a constraint on which configurations are physically realizable given the full boundary conditions of the experiment.
- Detection is not collapse; it is *boundary termination* — the phase field matching the boundary conditions imposed by the detector.
- Which-path information is not observation-dependent; it is *sector-selective* — the imposition of additional physical constraints that restrict the admissible set to a decohered sector.

The purpose of this paper is to prove the *Photon Non-Paradox Theorem* (PNPT), which states that within Phase Theory, LPST, and LST, no photon paradox exists. More precisely: every standard photon paradox can be traced to the illicit importation of object-trajectory predicates into a framework where such predicates are undefined. The theorem is constructive: it identifies the specific predicate responsible for each paradox and shows that removing it dissolves the paradox without altering any empirical prediction.

The paper is structured as follows. Section 2 reviews the phase-primitive framework, emphasizing the shared ontological and mathematical structure across the three theories. Section 3 presents six formal definitions that ground the theorem. Section 4 states the Photon Non-Paradox Theorem. Section 5 presents the full proof through four lemmas and a concluding argument. Section 6 derives four corollaries resolving each classical paradox. Section 7 applies the theorem to five canonical experiments. Section 8 compares the PNPT resolution with standard quantum interpretations.

Section 9 discusses scope, limitations, and open questions. Section 10 provides the compact, paper-ready statement of the theorem. Section 11 concludes.

2. Review of the Phase-Primitive Framework

This section presents the shared ontological and mathematical structure underlying Phase Theory [30], LPST [31], and LST [32], with emphasis on the features directly relevant to the Photon Non-Paradox Theorem. While the three theories differ in mathematical detail — Phase Theory employs a map to a compact target manifold, LPST employs a complex scalar field with internal polarization, and LST employs a substrate tensor field — they converge on a common conceptual architecture that we now describe.

2.1 Ontological Foundation

All three theories posit a single ontological primitive: a *phase-bearing substrate field*. The physical content of the universe is exhausted by the configuration of this field; there are no additional entities, forces, or background structures.

In *Phase Theory*, the substrate is specified by a continuous map $\Phi: M \rightarrow T$, where M is a connected topological base space (the domain of the field) and T is a compact target manifold (the space of phase values). The base space M is not identified with physical spacetime; spacetime emerges from the correlation structure of Φ . The target manifold T may be as simple as S^1 (yielding a single angular phase variable) or as rich as a higher-dimensional compact Lie group or homogeneous space.

In *LPST*, the substrate is a complex scalar field $L = A \exp(i\varphi)$ equipped with an internal polarization vector $\mathbf{u}$ defined on a label space Σ . The label space Σ provides the pre-geometric arena; physical spacetime emerges from the interference and correlation structure of L . The amplitude A and phase φ jointly encode all physical content, with the phase φ carrying the primary dynamical information.

In *LST*, the substrate is a tensor field $S_{\mu\nu}$ defined on an emergent manifold. The tensor structure accommodates both scalar (trace) and tensorial (traceless)

excitations, which are identified with matter and geometry respectively. The emergent manifold arises from the substrate's self-organizing dynamics.

Across all three theories, three structural commitments are shared: (i) spacetime is *emergent*, not a pre-existing background; (ii) matter consists of *stable topological defects* in the substrate field — knots, vortices, domain walls, skyrmions, or their higher-dimensional generalizations; (iii) geometry arises from the *correlation structure* of the substrate, in a manner structurally analogous to the holographic derivation of geometry from entanglement (cf. Ryu-Takayanagi [29], Maldacena [28]).

2.2 The Consistency Principle

The dynamical content of all three theories is governed by a single overarching principle: *physical configurations minimize a global phase-inconsistency functional*.

Let $\mathcal{I}[\Phi] \geq 0$ denote this functional. In Phase Theory, this is **Axiom 1** (the consistency axiom): the physical configuration Φ_{phys} minimizes $\mathcal{I}[\Phi]$ subject to topological and boundary constraints. In LPST, this is **Postulate 3** (the phase consistency principle): closed-loop phase integrals must satisfy quantization conditions, and physical configurations extremize the phase-consistency action. In LST, the dynamics arise from the *substrate action*, whose Euler-Lagrange equations select configurations of extremal consistency.

The minimization is *constrained*. Two classes of constraints are universal across the theories: (a) *topological sector constraints* — the configuration must lie in a specified topological sector, characterized by conserved topological charges (winding numbers, linking numbers, Chern numbers, holonomy classes); and (b) *coherence bounds* — the phase field has finite coherence length and time, bounding the domain over which consistency minimization operates.

This principle replaces the standard time-first Hamiltonian evolution picture. Instead of specifying an initial state and evolving it forward in time via a Hamiltonian or Lagrangian, the phase-primitive framework specifies the full set of boundary conditions (preparation, propagation geometry, detection) and solves for the admissible configurations that minimize I subject to constraints. Time ordering is

emergent, not primitive — a feature that will prove essential in the resolution of the delayed-choice paradox (Lemma 4).

2.3 Admissibility and Topological Sectors

The space of all conceivable field configurations is denoted $\{\Phi\}$. The *admissible* configurations form a subset $A \subset \{\Phi\}$, determined by the topological sector, coherence bounds, and boundary conditions of the experimental arrangement.

A is partitioned into *topological sectors* by homotopy invariants. Two configurations Φ_1, Φ_2 belong to the same sector if and only if they are connected by a continuous deformation (homotopy) that preserves the boundary conditions and does not cross a topological obstruction. The classification is given by the homotopy groups $\pi_n(T)$ of the target manifold (in Phase Theory), by the winding numbers of closed-loop phase integrals (in LPST), or by the defect charges of the substrate tensor field (in LST).

Transitions between topological sectors require *topological events*: pair creation and annihilation of defects, reconnection of vortex lines, or nucleation of domain walls. These events are the phase-primitive analogues of particle creation and annihilation. Within a given sector, continuous deformation preserves the sector label and all associated conserved quantities.

2.4 Coherence and Decoherence

All three theories impose coherence limitations on the substrate field. In Phase Theory, **Axiom 4** bounds the mutual information between subsystems by the area of their shared boundary — a holographic coherence bound structurally analogous to the Bekenstein bound [26] and the holographic principle [27]. In LPST and LST, coherence limitations arise dynamically from interaction with environmental degrees of freedom.

Decoherence, in the phase-primitive framework, is not collapse. It is the restriction of the admissible set A when environmental record constraints are imposed. When a subsystem becomes correlated with an environment in a manner that creates a stable record, the admissible configurations are restricted to those consistent with the record. This restriction is a physical process — it modifies A — but it does not

involve any non-unitary operation, any special role for consciousness, or any ontological discontinuity.

2.5 Detection and Measurement

In Phase Theory (Sections XVII–XVIII), measurement is identified with decoherence: no separate collapse postulate is needed or introduced. The Born rule emerges from phase averaging over the coherence domain. In LPST (Section 22), quantum mechanics is recovered as an effective description of the substrate dynamics; the Feynman path integral emerges from the phase-consistency principle as a sum over admissible configurations weighted by $\exp(i\mathbb{I}[\Phi]/\hbar)$. In LST (Section 20), entanglement is interpreted as substrate continuity — two subsystems are entangled if and only if their phase configurations are linked by a continuous substrate connection — and measurement is a local boundary condition imposed on the substrate field.

In all three theories: detection is a *boundary-matching event*, not a mysterious intervention by an observer. The detector occupies a region of the effective domain $\mathcal{M}$ and imposes a set of boundary conditions on Φ . The detection event occurs if and only if the admissible configuration Φ_{phys} satisfies these boundary conditions. This is analogous to a standing wave satisfying nodal conditions imposed by the walls of a cavity. No additional axiom of collapse, measurement, or observation is required or permitted.

3. Primitives and Definitions

We now formalize six definitions that provide the conceptual and mathematical foundation for the Photon Non-Paradox Theorem. Each definition is stated formally, followed by commentary clarifying its content, motivation, and relationship to the parent theories.

Definition 1 (Phase Substrate).

Let Φ denote the total physical state, represented as a phase field (or phase configuration) on a substrate domain M . The domain M serves as the effective arena for predictions; it may be a connected topological space (Phase Theory), a label space Σ (LPST), or an emergent manifold (LST). In Phase Theory, $\Phi: M \rightarrow T$ is a continuous map to a compact target manifold T . In LPST, the field is $L = A \exp(i\varphi)$ with internal polarization $\mathbf{u}$ on Σ . In LST, the field is a substrate tensor $S_{\mu\nu}$ on M . No pre-existing spacetime metric is assumed on M ; the metric, if present, is derived from the correlation structure of Φ .

Commentary. The domain M differs in its detailed character across the three theories, but its *functional role* is identical: it provides the arena over which the phase field is defined and over which the consistency functional is evaluated. The crucial point for the PNPT is that M is not spacetime in the usual sense — it does not come equipped with a Lorentzian metric, causal structure, or time-ordering. These features emerge from Φ . Consequently, notions such as "the photon was at position at time t " are not primitive statements in the theory; they are derived, approximate descriptions valid only within the emergent spacetime regime.

Definition 2 (Admissible Configurations).

Let $A \subset \{\Phi\}$ denote the set of admissible configurations, defined by three constraints:

*(i) **Topological admissibility:** The configuration lies in an allowed topological sector, specified by the homotopy class $[\Phi] \in \pi_n(T)$ (or the corresponding winding/defect class in LPST/LST).*

*(ii) **Coherence bounds:** The configuration satisfies the coherence limitations imposed by Axiom 4 (Phase Theory) or the dynamical decoherence mechanisms (LPST/LST). In particular, the phase field has finite coherence length ℓ_c and coherence time τ_c , and phase correlations decay beyond these scales.*

*(iii) **Boundary consistency:** The configuration is compatible with the*

preparation and apparatus constraints. The source, beam splitters, mirrors, and detectors define boundary conditions on Φ , and only configurations satisfying these boundary conditions are admissible.

Commentary. The admissible set A encodes the full physical content of the experimental arrangement. Changing any element of the apparatus — inserting a beam splitter, adding a which-path detector, modifying the source — changes A . This is a physical modification, not an epistemic update. Two experiments with different apparatus have different admissible sets, and therefore different physical predictions, regardless of what any observer knows or believes. The role of A in the PNPT is to make precise the distinction between "interference-admissible" and "which-path-admissible" configurations, which will be shown to occupy disjoint sectors.

Definition 3 (Phase Action / Consistency Functional).

Let $I[\Phi] \geq 0$ be a functional measuring global phase inconsistency (equivalently, negative log weight), such that physical realizations concentrate on configurations minimizing I subject to the admissibility constraints:

$$\Phi_{phys} \in \arg \min_{\Phi \in A} I[\Phi] \quad (1)$$

The functional I takes the following forms in the respective theories. In Phase Theory:

$$I[\Phi] = \int_M \rho(\Phi, D\Phi) d\mu, \quad \rho = \frac{1}{2} K_{ab} D_\mu \Phi^a D^\mu \Phi^b + V(\Phi) + \lambda \text{Tr}(F_{\mu\nu} F^{\mu\nu}) \quad (2)$$

In LPST, the loop-integral form:

$$I[\varphi] = \sum_{\text{closed loops}} \oint \nabla \varphi \cdot d\mathbf{l} - 2\pi n \quad (3)$$

In LST, the substrate action $S[S_{\mu\nu}]$ with nonlinear self-interaction terms.

All forms share: non-negativity, gauge invariance, locality of the integrand, and topological invariance of the minima.

Commentary. The consistency functional $I[\Phi]$ plays the role that the action plays in standard Lagrangian field theory, but with a crucial conceptual difference: it is interpreted not as generating time-evolution but as encoding global consistency constraints. The physical configuration is the one that minimizes total inconsistency given the full set of boundary conditions. This is a constraint-satisfaction picture, not a dynamical-evolution picture. The distinction is essential for the resolution of the delayed-choice paradox (Lemma 4): because I is evaluated over the full experimental geometry including all boundaries, there is no privileged temporal direction in the minimization.

Definition 4 (Photon Event).

A "photon" is not an object but an equivalence class of phase excitations. Formally: a photon is a label for a family $\Gamma \subset A$ of configurations whose restriction to the apparatus region produces a single quantized detection event. The equivalence class Γ is characterized by:

- (a) Γ consists of localized, metastable excitations $[\Phi]_v$ in a specified admissible sector.*
- (b) Members of Γ are capable of transporting a quantized exchange $\hbar\omega$ as a consequence of sector quantization and boundary matching.*
- (c) Two configurations $\Phi_1, \Phi_2 \in \Gamma$ are equivalent if they produce the same detection signature at all detectors: $E_k(\Phi_1) = E_k(\Phi_2)$ for all k .*

Commentary. This definition is the ontological core of the PNPT. In standard quantum field theory, a photon is a Fock-space excitation of the quantized electromagnetic field — a single-particle state $|1_{\mathbf{k},\lambda}\rangle$ characterized by wavevector $\mathbf{k}$ and polarization λ . In classical electrodynamics, a photon is (loosely) a wave packet of the electromagnetic field. In both cases, the photon is conceived as a *thing* — an entity with intrinsic properties that exist during propagation, independent of detection.

In the phase-primitive framework, the photon is defined *extensionally* — by what it does at boundaries — not *intensionally* — by what it "is" during propagation. Between source and detector, there is no photon-object; there is a phase field Φ constrained by admissibility. The "photon" label is a convenient shorthand for the equivalence class of admissible configurations that produce a single quantized detection event. Asking "which path did the photon take?" is asking for an intensional property of an entity that has only extensional definition. The question is not unanswerable; it is *unaskable* within the framework.

Definition 5 (Detection Event).

A detection event at detector D_k is a boundary satisfaction / termination condition:

$$E_k = \{\Phi \in A : \Phi \text{ satisfies } B_k \text{ and terminates coherence in } D_k\} \quad (4)$$

where B_k denotes the set of boundary conditions imposed by the physical detector D_k on the phase field Φ .

Commentary. Detection is not a mysterious "measurement" that collapses a wavefunction. It is the phase field matching a set of boundary conditions imposed by the physical detector. The detector provides *termination constraints*; the phase field either satisfies them or does not. This is analogous to a standing wave satisfying nodal conditions: the boundary shapes the solution, not the observation. The quantization of detection — why the detector registers a whole photon, never half a photon — follows from the topological quantization of the excitation class Γ : the topological invariant characterizing Γ is integer-valued, and the energy deposited at the detector is $\hbar\omega$ precisely because the topological charge transferred is quantized. The detector does not "choose" to absorb a whole quantum; the admissibility constraints *require* it.

Definition 6 (Which-Path Predicate).

A "which-path" statement W is a predicate on Φ requiring that the configuration factorize into mutually exclusive subdomains consistent with a definite path record:

$$W = \{\Phi \in A : \exists \text{ stable record } r \in \{1, 2\} \text{ selecting path subdomain } M_r \subset M\} \quad (5)$$

Commentary. The which-path predicate W is not a statement about observer knowledge; it is a *physical constraint*. A which-path device — a polarization marker, a micro-maser cavity, a quantum dot — imposes additional boundary conditions that constrain Φ to configurations possessing a stable environmental record distinguishing the path subdomains. This is a real physical modification of the admissible set A , not an epistemic update. When a which-path device is inserted, the admissible set changes from A to $A_W = A \cap W$. The resulting restriction is what eliminates interference, as Lemma 2 will establish.

4. The Photon Non-Paradox Theorem

Theorem 1 (Photon Non-Paradox Theorem).

In Phase Theory, LPST, and LST, under Definitions 1-6: For any experimental arrangement E involving preparation P , propagation region R , and measurement boundary conditions $\{B_k\}$, there exists no logically consistent set of predicates that simultaneously requires:

- (1) Object-like photon trajectory identity — the photon traverses a definite, single path through R ;*
- (2) Phase-coherent global admissibility — the admissible set A includes configurations constrained by boundary conditions across the full propagation region R ;*
- (3) Detection as a boundary termination event — detection E_k is the satisfaction of boundary conditions B_k , not a dynamical collapse;*

while preserving the admissibility constraints A. Therefore, any "photon paradox" is a contradiction introduced by adding non-primitive object predicates to a phase-primitive theory, not a contradiction in the theory's physical content.

Discussion. The Photon Non-Paradox Theorem asserts a precise structural claim: the five classical photon paradoxes enumerated in Section 1 are not features of nature, not features of quantum mechanics per se, and not features of the phase-primitive theories. They are features of a *mixed ontology* — an ontology that attempts to combine object-trajectory predicates (the photon is a thing that goes somewhere) with phase-admissibility predicates (interference is global, detection is boundary-matching). This mixed ontology is internally inconsistent, and the inconsistency manifests as "paradoxes."

The theorem does *not* claim that quantum mechanics is wrong. It does not claim that interference patterns are unexplained. It does not claim that photons do not exist. The empirical content of quantum optics is entirely preserved: interference patterns, complementarity inequalities, delayed-choice correlations, and detection statistics are all reproduced within the phase-primitive framework. What is eliminated is the *paradoxical* character of these phenomena — the sense that nature is doing something contradictory or mysterious.

The theorem *does* claim that the concept "photon paradox" is a *category error*: it arises from applying predicates appropriate to one ontological framework (objects with trajectories) within a different ontological framework (phase fields with admissibility constraints). This is analogous to the "paradox" of asking whether a poem is heavier than a symphony — the question mixes predicates from incompatible categories. In the phase-primitive framework, asking "which path did the photon take?" is not a question with a hidden answer; it is a question that presupposes an entity (a photon-object) that the framework does not contain.

The proof of the theorem is constructive: it proceeds through four lemmas, each of which identifies the specific object-predicate responsible for one class of paradox

and shows that its removal dissolves the paradox without altering any empirical prediction.

5. Proof

We establish the theorem through four lemmas, each addressing a specific aspect of the photon-paradox structure. The lemmas are logically independent but collectively exhaust the paradox space. The proof concludes with a reductio ad absurdum argument synthesizing all four lemmas.

5.1 Lemma 1: No Photon-Object Identity Map

Lemma 1 (No Photon-Object Identity Map).

There is no injective map $f: \Gamma \rightarrow \text{Trajectories}$ that assigns each admissible "photon" phase class Γ a unique classical trajectory while preserving A , because admissibility is global and generally non-factorizable across R .

Proof.

Consider an interferometric geometry: a Mach-Zehnder interferometer with beam splitters BS_1 and BS_2 , mirrors M_1 and M_2 , and detectors D_1 and D_2 . The propagation region R consists of two arms, arm 1 ($BS_1 \rightarrow M_1 \rightarrow BS_2$) and arm 2 ($BS_1 \rightarrow M_2 \rightarrow BS_2$).

The admissible configurations A_{int} for this geometry are constrained by boundary conditions on both arms simultaneously: the preparation conditions at BS_1 , the reflection conditions at M_1 and M_2 , the recombination conditions at BS_2 , and the detection conditions at D_1 and D_2 . The consistency functional $I[\Phi]$ couples the phase field across the entire propagation region R , because the boundary conditions at BS_2 involve phase differences between the two arms.

Now suppose, for contradiction, that an injective map f exists. Then each

admissible configuration $\Phi \in A_{\text{int}}$ is assigned a unique trajectory — either arm 1 or arm 2. Let $A_1 = f^{-1}(\text{arm 1}) \cap A$ and $A_2 = f^{-1}(\text{arm 2}) \cap A$ denote the trajectory-restricted subsets. For any configuration $\Phi_1 \in A_1$, the phase field is restricted to arm 1: $\Phi_1|_{\text{arm 2}}$ is unconstrained or trivial. Similarly for $\Phi_2 \in A_2$.

However, the minimizer Φ_{phys} of $I[\Phi]$ over the full admissible set A_{int} generally satisfies:

$$I[\Phi_{\text{phys}}] < \min(\inf_{\Phi \in A_1} I[\Phi], \inf_{\Phi \in A_2} I[\Phi]) \quad (6)$$

because cross-arm coherence terms — specifically, the phase-linking constraints between the two arms at BS_2 — lower the total inconsistency. The consistency functional evaluated at the output beam splitter includes a term proportional to:

$$|\Phi_{\text{arm1}}(BS_2) - \Phi_{\text{arm2}}(BS_2) e^{i\delta}|^2 \quad (7)$$

where δ is the accumulated phase difference between the arms. When both arms are populated, this cross-term can be minimized (potentially to zero for appropriate δ), yielding a lower value of I than any single-arm configuration can achieve. Therefore, trajectory-restricted configurations A_1 and A_2 fail admissibility in the global sense: they have strictly higher inconsistency and are not physical minimizers.

Since $A_1 \cup A_2$ does not contain Φ_{phys} , the map f cannot be injective while preserving membership in A_{int} . The assumption of an injective trajectory map is contradicted.

■

Discussion. Lemma 1 establishes that the failure of trajectory assignment is not epistemic but ontological. It is not that we *do not know* which path the photon took; it is that the concept of "which path" is *undefined* for configurations in A_{int} . The phase field does not factorize across arms in the interference-admissible sector. Any attempt to force a factorization produces a configuration with strictly higher inconsistency — a configuration that is not physically realized. The photon-object, in

the sense of an entity traveling a definite path, simply does not exist within the phase-primitive description of an interference experiment.

5.2 Lemma 2: Which-Path Records Incompatible with Interference Sector

Lemma 2 (Which-Path / Interference Sector Exclusivity).

If an apparatus introduces stable which-path records W (Definition 6), then the admissible set collapses to a decohered sector A_W in which cross-arm phase-linking constraints are disallowed by coherence bounds:

$$A_W \cap A_{\text{int}} = \emptyset \quad (8)$$

where A_{int} denotes the interference-admissible sector.

Proof.

A which-path device introduces a physical coupling between the photon-carrying phase mode and an environmental degree of freedom (the record). In Phase Theory terms, this expands the effective configuration space to include the record variable $r \in \{1, 2\}$, labeling path 1 or path 2. The admissible set with records is:

$$A_W = \{\Phi \in A : \exists \text{ stable } r \text{ such that } \Phi|_{\text{path } i} \text{ is correlated with } r = i\} \quad (9)$$

For W to constitute a *stable* record, the mutual information between r and the path must exceed the decoherence threshold set by the coherence limitation (Axiom 4 in Phase Theory, or the corresponding dynamical decoherence threshold in LPST/LST). Specifically, the record states $|r = 1\rangle$ and $|r = 2\rangle$ must be macroscopically distinguishable, meaning their overlap is negligible:

$$\langle r = 1 | r = 2 \rangle \approx 0 \quad (10)$$

Now consider the off-diagonal coherence terms that define membership in A_{int} .

These are the cross-arm phase-linking constraints — the overlap integral between path-1 and path-2 restricted configurations, weighted by the record states:

$$\int \Phi_1^* \Phi_2 \langle r = 1 \mid r = 2 \rangle d\mu = 0 \quad (11)$$

by equation (10). But these off-diagonal terms are precisely the cross-arm phase-linking constraints that define membership in A_{int} . A configuration belongs to A_{int} if and only if the cross-arm coherence terms are non-vanishing — this is what produces the interference pattern. Since the which-path record forces these terms to vanish, we have:

$$\Phi \in A_W \implies \text{cross-arm coherence} = 0 \implies \Phi \notin A_{\text{int}} \quad (12)$$

Therefore $A_W \cap A_{\text{int}} = \emptyset$. ■

Discussion. Lemma 2 dissolves the paradox of complementarity. You do not "destroy interference by observing." You physically *select a different admissible sector* by adding record constraints. The interference pattern is not "there but hidden"; the configurations that produce it are literally not in the admissible set once which-path records exist. This is a physical statement, not an epistemic one: the which-path device modifies the boundary conditions on Φ , and therefore modifies A , and therefore modifies the physical predictions.

Connection to the Englert-Greenberger duality relation. The quantitative complementarity inequality $D^2 + V^2 \leq 1$, where D is path distinguishability and V is fringe visibility [9, 16], receives a natural derivation within the PNPT framework. The distinguishability D measures the degree to which W -type constraints have been imposed — the extent to which the record states are orthogonal. The visibility V measures the degree to which A_{int} -type configurations contribute to the detection statistics — the magnitude of the cross-arm coherence terms. Their complementarity is a direct consequence of $A_W \cap A_{\text{int}} = \emptyset$: as D increases (the record becomes more

stable), the overlap with A_{int} decreases, and V must decrease. The inequality $D^2 + V^2 \leq 1$ is the quantitative expression of sector exclusivity.

5.3 Lemma 3: Detection is Boundary Termination, Not Propagation Collapse

Lemma 3 (Detection as Boundary Termination).

Detection E_k is a boundary condition satisfaction event: Φ must match B_k . There is no additional axiom of collapse; post-detection states are conditioned on E_k :

$$\Phi_{\text{post}} = \Phi / E_k \quad (13)$$

This conditioning is not a physical discontinuity in Φ 's ontology; it is a constraint update reflecting that coherence cannot extend past the termination boundary.

Proof.

In the phase-primitive framework, the detector D_k occupies a region of M and imposes boundary conditions B_k on the phase field. The detection event E_k is the subset of configurations in A that satisfy these boundary conditions (Definition 5).

Prior to detection, Φ_{phys} is determined by minimizing $I[\Phi]$ over all of A with the preparation and propagation constraints. After detection, Φ_{phys} is determined by minimizing $I[\Phi]$ over the subset E_k . The transition from A to E_k is a *restriction of the constraint set*, not a dynamical process acting on Φ . Formally:

$$\Phi_{\text{pre}} \in \arg \min_{\Phi \in A} I[\Phi] \rightarrow \Phi_{\text{post}} \in \arg \min_{\Phi \in E_k} I[\Phi] \quad (14)$$

There is no "collapse" operator, no non-unitary evolution, and no instantaneous action at a distance. The phase field at the detector simply matches the detector's boundary conditions, just as a standing wave matches its nodal constraints.

The *quantization of detection* — why the detector registers a whole photon, never half a photon — follows from the topological quantization of the excitation class Γ (Definition 4). The equivalence class Γ is characterized by a topological invariant (winding number, Chern number) that is integer-valued. The energy deposited at the detector is $\hbar\omega$ precisely because the topological charge transferred is quantized:

$$\Delta Q_{top} = n \in \mathbb{Z}, \Delta E = n\hbar\omega \quad (15)$$

This is boundary matching, not mystical collapse. The detector's boundary conditions are compatible only with integer topological charge transfer, so no fractional detection event is admissible. ■

Discussion. Lemma 3 eliminates the measurement problem. In the Copenhagen interpretation, collapse is a *physical process* — an instantaneous, non-unitary change of state triggered by measurement. In the Many-Worlds interpretation, collapse is avoided at the cost of universal branching — every measurement spawns a new branch of reality. In Bohmian mechanics, collapse is avoided by introducing hidden trajectories guided by a pilot wave. In the phase-primitive framework, *none of these auxiliary mechanisms are needed*. Detection is boundary termination: the phase field satisfies the boundary conditions imposed by the detector, the topological charge is transferred in quantized units, and the admissible set is updated to reflect the new constraint. This is as physically transparent as a standing wave in a vibrating string adjusting to a new boundary condition when a clamp is applied.

5.4 Lemma 4: Delayed-Choice is Boundary Reclassification, Not Retrocausality

Lemma 4 (Delayed-Choice as Boundary Reclassification).

Changing the measurement context after preparation alters the set of admissible boundaries $\{B_k\}$, hence changes A for the experiment as defined. The theory evaluates Φ_{phys} by minimizing I subject to the actual boundaries. "Paradoxical" narratives arise only if one assumes the photon carried a pre-existing object property (e.g., "went left") independent of boundaries — an assumption rejected by Definition 4.

Proof.

Consider Wheeler's delayed-choice experiment [5, 6]. A photon enters a Mach-Zehnder interferometer. After the photon has "passed" the first beam splitter BS_1 (in the conventional temporal narrative), the experimenter chooses whether to insert or remove the second beam splitter BS_2 :

- **BS₂ inserted:** The apparatus boundary conditions include phase-recombination constraints at BS_2 . The admissible set is A_{int} — the interference-admissible sector. Interference is observed.
- **BS₂ removed:** The apparatus boundary conditions include only the individual detector constraints at D_1 and D_2 , without recombination. The admissible set is A_w — the which-path sector. Which-path information is obtained.

The conventional paradox asserts that the photon must have "decided" to be a wave or a particle before the experimenter's choice was made, implying retrocausal influence.

In the PNPT framework: there is no photon that "decided" anything. The phase field Φ is constrained by the *actual* boundary conditions of the entire apparatus, including the final measurement configuration. The consistency functional $I[\Phi]$ is minimized over the admissible set A defined by the actual experimental arrangement — all of it, simultaneously.

Formally, let E_{ins} denote the experiment with BS_2 inserted and E_{rem} denote the

experiment with BS_2 removed. These are *different experiments* with different boundary conditions:

$$A(E_{ins}) = A_{int}, A(E_{rem}) = A_W \quad (16)$$

The phase-consistency minimization is a *global constraint satisfaction* over the entire experimental geometry, evaluated at the actual boundaries. There is no temporal ordering in the fundamental description — time is emergent from update ordering (Phase Theory Axiom 2, LPST Postulate 2). The phase field does not "first" pass BS_1 , "then" travel, "then" encounter BS_2 . The admissibility constraints are imposed simultaneously over the full domain M .

The appearance of retrocausality arises *only* if one insists on a temporal narrative in which the photon first passes BS_1 , then travels through the arms, then encounters BS_2 . This narrative imports object-trajectory ontology: there is a *thing* (the photon) that *passes through* (traverses a trajectory) in a temporal sequence. In the phase-primitive framework, this narrative has no fundamental status. The experiment is specified by its full set of boundary conditions, and the physical configuration is the global minimizer of $I[\Phi]$ subject to those conditions.

■

Discussion. Lemma 4 provides a natural home for the Aharonov-Bergmann-Lebowitz (ABL) time-symmetric formulation of quantum mechanics [15] and the two-state vector formalism. In the ABL framework, the quantum state is specified by both an initial (pre-selected) state and a final (post-selected) state, and probabilities are computed using both. In the PNPT framework, the "pre-selection" corresponds to the preparation boundary conditions (source at BS_1) and the "post-selection" corresponds to the measurement boundary conditions (detector configuration). Both are simply components of the full boundary set $\{B_k\}$ that defines A . The two-state vector formalism is not an interpretive add-on; it is the natural description of phase-consistency minimization over a bounded domain.

The experimental results of Jacques *et al.* (2007) [11] — in which the choice of measurement configuration was made at spacelike separation from the photon's

entry into the interferometer, using a quantum random number generator — are naturally explained within the PNPT framework. The phase field's admissibility is determined by the actual apparatus geometry at the time the consistency functional is evaluated (i.e., the actual physical boundary conditions that exist when the experiment is complete), not by a temporal narrative about when the photon "was" in the interferometer. No retrocausation, no faster-than-light signaling, and no observer-dependent reality are invoked.

5.5 Conclusion of the Proof

Proof of Theorem 1 (Conclusion).

Assume, for contradiction, that a "photon paradox" exists within Phase Theory / LPST / LST. Then one must jointly assert the following three predicates within the phase-primitive framework:

(O) The photon is an object with a definite path predicate W — it traverses a specific trajectory through the propagation region R .

(I) Interference sector admissibility — the admissible set A_{int} includes configurations with non-vanishing cross-arm phase coherence, producing interference.

(D) Detection as termination conditioning — detection E_k is the satisfaction of boundary conditions B_k , without collapse.

By **Lemma 1**, predicate (O) requires an injective trajectory map $f: \Gamma \rightarrow \text{Trajectories}$. But no such map exists while preserving membership in A_{int} , because admissibility is global and the consistency functional couples both arms. Therefore (O) and (I) are jointly incompatible within A .

By **Lemma 2**, if (O) is enforced through a which-path record W , the admissible set collapses to A_W , and $A_W \cap A_{\text{int}} = \emptyset$. Therefore (O) and (I) select disjoint admissible sectors: they cannot both describe the same physical configuration.

By **Lemma 3**, predicate (D) is independently well-defined: detection is boundary

termination, requiring no collapse and introducing no paradox.

By **Lemma 4**, any temporal narrative suggesting that the photon's past behavior must be retroactively determined presupposes (O) — that the photon carried a definite trajectory property during propagation — which is precisely what Lemma 1 rules out.

Therefore, the paradox arises *exclusively* from the addition of (O) — the object-trajectory predicate — to a framework in which (O) is not a primitive and is, in fact, incompatible with the admissibility structure. The paradox does not exist within the theory; it exists only in a *mixed ontology* that illegitimately combines phase-primitive and object-primitive predicates.

Since the argument is independent of the specific experimental arrangement E (the lemmas apply to any interferometric, which-path, delayed-choice, or detection geometry), the theorem holds universally for all standard photon-paradox scenarios within Phase Theory, LPST, and LST.



6. Corollaries

The Photon Non-Paradox Theorem yields four immediate corollaries, each resolving a specific classical paradox.

Corollary 1 (Wave-Particle Duality is a False Dichotomy).

"Wave" and "particle" are not competing ontologies for the photon. "Wave" denotes the phase coherence constraints on A — the global admissibility structure that produces interference patterns. "Particle" denotes the boundary termination quantization at $\{B_k\}$ — the discrete, integer-valued topological charge transfer at detectors. These are not mutually exclusive properties of a thing; they are different aspects of admissible phase dynamics. The wave-particle "duality" dissolves because both descriptions are partial projections of a

single phase-admissibility structure.

Discussion. A "wave" description captures the global coherence properties of configurations in A_{int} : wavelength, phase, superposition, and interference. A "particle" description captures the quantized boundary-matching properties of detection events E_k : discreteness, localization, and indivisibility. These are no more contradictory than the length and width of a rectangle. They describe different aspects of the same underlying reality — the phase substrate — projected onto different observational contexts. Bohr's complementarity principle [2] is not wrong but is revealed to be incomplete: it correctly identifies the mutual exclusivity of wave and particle *descriptions* but misidentifies the source as a fundamental limitation of nature rather than a limitation of object-trajectory ontology.

Corollary 2 (Self-Interference is Global Admissibility, Not Self-Action).

The statement that a "single photon interferes with itself" (Dirac, 1930 [3]) is re-described as: configurations in A_{int} are globally constrained by the full geometry of the apparatus. There is no indivisible object that is "in two places at once." The interference pattern at the output is a property of the admissible set A_{int} for the given apparatus geometry, not a property of a particle performing an inexplicable act. Each detection event samples from A_{int} according to the consistency-weighted distribution.

Discussion. Dirac's dictum is empirically correct — single-photon interference is a real phenomenon, demonstrated from Taylor (1909) [33] through Grangier *et al.* (1986) [34] and beyond. But the ontological gloss — "a single particle interferes with itself" — imports object ontology. In the phase-primitive framework, there is no particle to be in two places. The phase field is defined over the entire apparatus domain M , and the admissible configurations in A_{int} are constrained by the boundary conditions at *both* slits (or both arms). The interference pattern is a collective property of the admissible set, not an action performed by an individual entity. This

is no more mysterious than the fact that a vibrating drumhead's pattern depends on the shape of the entire drum, not on any individual point of the membrane.

Corollary 3 (Complementarity is Sector Exclusivity).

Bohr's complementarity principle is re-derived as a mathematical consequence of the sector exclusivity established in Lemma 2:

$$A_W \cap A_{\text{int}} = \emptyset \quad (17)$$

Which-path and interference correspond to disjoint admissible sectors, not mutually exclusive "properties" of a thing. This is the phase-theoretic derivation of the Englert-Greenberger duality relation $D^2 + V^2 \leq 1$ [9, 16].

Discussion. Complementarity is often presented as a brute empirical fact or a fundamental principle of nature. In the PNPT framework, it is a *theorem* — a derived consequence of the structure of the admissible set. The derivation is transparent: which-path records impose constraints that are logically incompatible with the cross-arm coherence required for interference. The complementarity inequality $D^2 + V^2 \leq 1$ is the quantitative measure of this logical incompatibility. In this light, complementarity is demystified: it is not a deep metaphysical principle about the limits of human knowledge but a mathematical consequence of sector structure in phase space.

Corollary 4 (No Collapse Postulate Needed).

Conditioning on the detection event E_k (Definition 5, Lemma 3) suffices to produce all empirical predictions of quantum measurement theory, including the Born rule, state update, and detection statistics. Collapse is an interpretive overlay from non-primitive wavefunction ontology. Within the phase-primitive framework: the Born rule follows from phase averaging (Phase Theory Sec. XVII.2 [30]), the path integral emerges from phase consistency (LPST Sec. 22.1 [31]), and detection statistics follow from boundary-matching multiplicity (the

number of admissible configurations in Γ satisfying B_k).

Discussion. The collapse postulate has been the most criticized element of quantum mechanics since von Neumann's formalization [20]. It introduces a sharp dichotomy between "unitary evolution" (when no measurement occurs) and "non-unitary collapse" (when a measurement occurs) that has never been satisfactorily delineated. The PNPT eliminates this dichotomy entirely. There is only one kind of physical process: the phase field minimizing its consistency functional subject to boundary constraints. Detection adds a boundary constraint; it does not trigger a fundamentally different kind of physics. The resulting framework is ontologically uniform — one primitive, one dynamical principle, one kind of event — and empirically equivalent to standard quantum mechanics in all currently accessible regimes.

7. Application to Canonical Experiments

We now apply the PNPT to five famous experiments in quantum optics, demonstrating in each case how the paradoxical interpretation dissolves within the phase-primitive framework.

7.1 The Double-Slit Experiment

Setup. A source emits photons (at arbitrarily low intensity) toward a barrier with two slits, S_1 and S_2 , separated by distance d . A detection screen is placed at distance L behind the barrier. When both slits are open, an interference pattern builds up on the screen, one detection at a time. When one slit is blocked, the interference pattern disappears.

PNPT analysis. The admissible set A_{int} for the two-slit geometry contains configurations where the phase field threads both slits and satisfies boundary conditions at the screen. The consistency functional $I[\Phi]$ couples the phase field across the full propagation region, including both slit apertures. The boundary conditions at the screen detectors involve interference between contributions from

both slits — specifically, phase differences of the form $\Delta\phi = (2\pi/\lambda)(d \sin \theta)$, where θ is the angle from the central axis.

No single-slit restriction is admissible when both slits are open (Lemma 1). The configurations in A_{int} are globally constrained by the two-slit geometry; they do not factorize into "path through S_1 " and "path through S_2 " subsets. The interference pattern arises because the consistency-minimizing configurations have non-vanishing cross-slit coherence, leading to constructive and destructive modulation at the screen.

Adding a which-path detector at one slit — for example, a polarization rotator that marks which slit the light passed through — imposes the which-path predicate W (Definition 6). This restricts the admissible set to A_W (Lemma 2), in which cross-slit coherence vanishes. The interference pattern disappears — not because "observation disturbed something" but because the admissible sector changed. The physical boundary conditions of the apparatus are different, and therefore the physical predictions are different.

Detection at screen position x_k is boundary termination E_k (Lemma 3). The detector at position x_k imposes boundary conditions B_k on the phase field, and the detection event occurs if Φ_{phys} satisfies B_k . The probability of detection at x_k is proportional to the consistency-weighted measure of configurations in A_{int} that satisfy B_k , which reproduces the standard quantum-mechanical prediction $|\psi(x_k)|^2$.

7.2 The Mach-Zehnder Interferometer

Setup. A single photon enters a balanced Mach-Zehnder interferometer consisting of a 50:50 beam splitter BS_1 , mirrors M_1 and M_2 , a second 50:50 beam splitter BS_2 , and detectors D_1 and D_2 at the two output ports. With equal arm lengths and no phase shift, standard quantum mechanics predicts deterministic output: 100% of detections at D_1 (constructive interference) and 0% at D_2 (destructive interference).

PNPT analysis. The admissible set A_{int} for the balanced Mach-Zehnder interferometer contains configurations where the phase field threads both arms and satisfies the recombination boundary conditions at BS_2 . The consistency functional $I[\Phi]$ includes cross-arm coherence terms at BS_2 (cf. equation (7)). For balanced arms

(equal optical path lengths), the phase difference $\delta = 0$, and the minimizer Φ_{phys} satisfies:

$$\Phi_{\text{phys}}(D_1) = \Phi_{\text{arm1}} + \Phi_{\text{arm2}} \text{ (constructive)}, \Phi_{\text{phys}}(D_2) = \Phi_{\text{arm1}} - \Phi_{\text{arm2}} = 0 \text{ (destructive)} \quad (18)$$

Therefore E_1 is the only non-empty detection set: the consistency-minimizing configuration satisfies B_1 (detector D_1 fires) and does not satisfy B_2 (detector D_2 does not fire). This reproduces the deterministic quantum prediction without invoking any notion of the photon "choosing a path" or "being in both arms simultaneously."

Removing BS_2 . If BS_2 is removed, the apparatus boundary conditions change: there are no recombination constraints, only individual detector constraints at D_1 and D_2 . The admissible set transitions from A_{int} to A_{W} : each detection event is associated with a definite arm (D_1 registers photons from arm 1, D_2 from arm 2), with 50:50 probability. The which-path information is obtained because the admissible sector has changed.

7.3 Wheeler's Delayed-Choice Experiment

Setup. The Mach-Zehnder interferometer is modified so that the decision to insert or remove BS_2 is made *after* the photon has entered the interferometer (passed BS_1). Experimentally realized by Jacques *et al.* (2007) [11] using an electro-optic modulator switched by a quantum random number generator at spacelike separation from the photon's entry.

PNPT analysis (Lemma 4). The phase field's admissibility is determined by the actual boundary conditions of the complete apparatus — including the final configuration of BS_2 . If BS_2 is inserted (regardless of when the decision is made), the boundary conditions define A_{int} , and interference is observed. If BS_2 is removed, the boundary conditions define A_{W} , and which-path information is obtained.

There is no retrocausality. The phase-consistency minimization is not a forward-propagating signal that must "decide" at BS_1 whether to be a wave or a particle. It is a global constraint satisfaction over the entire domain M , evaluated with respect to the actual boundaries $\{B_k\}$ that characterize the completed experiment. The

temporal narrative ("the photon already passed BS_1 ") is an object-trajectory narrative that has no fundamental status in the phase-primitive framework, where time is emergent.

The experimental results of Jacques *et al.* [11] and Manning *et al.* [38] confirm the predictions: interference is observed when BS_2 is present, which-path correlations when it is absent, regardless of the timing of the choice. These results are precisely what the PNPT predicts, without any retrocausal mechanism.

7.4 The Quantum Eraser

Setup. In the quantum eraser experiment (Scully and Drühl, 1982 [7]; Kim *et al.*, 2000 [10]), entangled photon pairs are produced. The "signal" photon passes through a double slit with which-path markers (e.g., orthogonal polarizations at each slit). The "idler" photon is directed to an eraser apparatus where a measurement can either preserve or erase the which-path information. When the idler measurement erases the which-path information, interference is recovered in the coincidence counts of the signal photon. When the idler measurement preserves the which-path information, no interference is observed.

PNPT analysis. The which-path markers impose the predicate W , restricting the signal photon's admissible set to A_w . However, the signal and idler photons are entangled — in phase-primitive terms, their phase configurations are linked by substrate continuity (LST) or correlated boundary conditions (Phase Theory, LPST).

When the idler measurement *erases* the record — for example, by projecting the idler onto a basis that does not distinguish the which-path information — the constraint W is removed for the relevant correlation subspace. The admissible set for the signal photon, conditioned on the idler's erasure outcome, is restored from A_w back toward A_{int} . The interference pattern reappears in the coincidence counts.

Crucially, this is not retrocausation. The interference does not reappear in the *total* signal-photon detection pattern (which always shows no interference when which-path markers are present). It appears only in the *conditioned* subset of signal detections that are correlated with a specific idler outcome. The conditioning selects a subset of admissible configurations — those consistent with the erasure boundary

condition on the idler — and this subset belongs to A_{int} . The PNPT interpretation is straightforward: the boundary constraints defining admissibility include the idler measurement, and modifying those constraints modifies the admissible sector.

7.5 Elitzur-Vaidman Interaction-Free Measurement

Setup. A Mach-Zehnder interferometer is arranged for destructive interference at D_2 (D_2 never fires with balanced arms). A bomb (an ultra-sensitive absorber) is placed on one arm (say, arm 2). If the bomb is present: (a) with probability $\frac{1}{2}$, the bomb absorbs the photon (bomb explodes, photon detected at bomb); (b) with probability $\frac{1}{4}$, D_1 fires (consistent with no interference — inconclusive); (c) with probability $\frac{1}{4}$, D_2 fires — but D_2 firing is impossible with balanced, unobstructed arms [14]. Therefore, a D_2 detection reveals the bomb's presence *without the photon having interacted with the bomb*. This has been called "interaction-free measurement."

PNPT analysis. The bomb's presence on arm 2 modifies the admissible set. Configurations that thread arm 2 are subject to an additional boundary condition: the bomb's absorber imposes a termination constraint B_{bomb} . Configurations that satisfy B_{bomb} terminate at the bomb (detection by bomb, probability $\frac{1}{2}$). Surviving configurations — those that do not terminate at the bomb — are restricted to the subset of A that does not thread arm 2 with sufficient amplitude to trigger absorption.

This restriction changes the output statistics at D_1 and D_2 . Without the bomb, A_{int} produces destructive interference at D_2 (D_2 never fires). With the bomb, the admissible set is modified: the bomb removes a subset of configurations from A_{int} (those that would have threaded arm 2 and been absorbed), and the remaining configurations no longer produce perfect destructive interference at D_2 . Therefore D_2 can fire.

No mysterious "interaction-free knowledge transfer" is needed. The change in admissible set is *physical*: the bomb's presence modifies the boundary conditions on M , which modifies A , which modifies the detection statistics. The photon did not "avoid the bomb and learn about it anyway." The phase field's admissibility was globally altered by the bomb's presence as a boundary condition, even for

configurations that did not terminate at the bomb. This is the hallmark of phase-primitive ontology: admissibility is global, not local.

8. Relationship to Quantum Interpretations

The Photon Non-Paradox Theorem provides a unified resolution of the standard photon paradoxes. It is instructive to compare this resolution systematically with those offered by the major interpretive frameworks of quantum mechanics.

8.1 Copenhagen Interpretation

The Copenhagen interpretation, associated with Bohr, Heisenberg, and (in its formalized version) von Neumann [2, 20], accepts complementarity as a fundamental principle and invokes the collapse postulate to account for measurement. The PNPT intersects with Copenhagen in accepting the empirical content of complementarity but diverges sharply in its treatment of collapse and the status of complementarity. The PNPT *derives* complementarity as a theorem (Corollary 3, from Lemma 2) rather than postulating it, and *eliminates* collapse entirely (Corollary 4, from Lemma 3). The "measurement problem" — the question of what constitutes a measurement and why it triggers a special physical process — does not arise in the PNPT framework because detection is boundary termination, not a special physical process requiring a separate postulate.

8.2 Many-Worlds Interpretation

The Many-Worlds interpretation (Everett, 1957 [22]) avoids collapse by postulating that the universal wavefunction never collapses; instead, every measurement causes a branching of the universe into non-communicating branches. The PNPT shares the commitment to avoiding collapse but achieves this without branching. There is no universal wavefunction to split — there is a phase field Φ constrained by admissibility. Different measurement boundary conditions define different admissible sets A , producing different physical predictions, but these are properties of different experimental arrangements, not parallel universes. The ontological

economy of the PNPT (one phase field, one set of boundary conditions, one outcome) is substantially greater than that of Many-Worlds.

8.3 Bohmian Mechanics

Bohmian mechanics (Bohm, 1952 [21]) restores definite trajectories by introducing a guiding wave (the pilot wave) that determines the particle's actual path. The PNPT takes the opposite approach: rather than restoring trajectories, it *eliminates them* (Lemma 1). The Bohmian guiding wave is, in some respects, analogous to the admissibility constraints of the phase-primitive framework — both provide a global constraint on outcomes. However, Bohmian mechanics retains the photon as an object (with a definite position at all times) and introduces the guiding wave as an additional ontological element. The PNPT has no hidden variables, no definite positions during propagation, and no guiding wave. The phase field itself is the sole primitive.

8.4 QBism and Relational Quantum Mechanics

QBism (Fuchs, 2010 [23]) treats quantum states as personal probability assignments reflecting an agent's beliefs, not objective features of reality. Relational quantum mechanics (Rovelli, 1996 [24]) treats quantum states as relational — defined only with respect to a specific observer-system pair. The PNPT agrees with both approaches that the "quantum state" (the wavefunction or density matrix) is not a property of the photon-object — there is no such object. However, the PNPT goes further: the admissibility constraints that determine physical predictions are *objective features of the phase substrate*, not subjective beliefs (contra QBism) and not observer-relative (contra relational QM). Two experimenters analyzing the same apparatus with the same boundary conditions will compute the same admissible set A and make the same predictions. The objectivity resides in the phase substrate and its boundary conditions, not in a wavefunction.

8.5 Decoherence Program

The PNPT is most closely aligned with the decoherence program initiated by Zeh (1970) [25] and developed extensively by Zurek (2003) [12] and Joos *et al.* (2003)

[13]. Lemma 2 can be seen as a formal derivation of environment-induced decoherence within phase-primitive ontology: the which-path device creates environmental correlations that suppress off-diagonal coherence terms, selecting a decohered sector. The key difference is that decoherence within standard quantum mechanics still requires an *interpretation* of the residual density matrix. After decoherence, the density matrix is diagonal in the pointer basis, but one must still explain why a *single* outcome is observed — the "and/or" problem. In the PNPT framework, the density matrix is an *effective description* of phase-averaging over the coherence domain. Single outcomes arise from boundary termination (Lemma 3): the detector's boundary conditions select a single admissible configuration from the decohered set. No additional interpretive step is required.

9. Discussion

Scope and generality. The Photon Non-Paradox Theorem, as stated and proved, applies to all optical interference experiments: double-slit, Mach-Zehnder, Fabry-Pérot, Sagnac, and their variants, as well as delayed-choice, quantum-eraser, and interaction-free measurement configurations. The theorem extends naturally to *matter-wave interference* — electrons (Tonomura *et al.*, 1989 [35]), neutrons (Greenberger and Yasin, 1988 [16]), and large molecules (Arndt *et al.*, 1999 [36]) — because the same phase-admissibility structure applies to all excitations of the substrate. In the phase-primitive framework, electrons, neutrons, and molecules are topological defects of the same substrate field Φ ; their wave-like behavior reflects the global admissibility constraints on Φ , just as for photons.

Experimental implications. The PNPT does not predict different experimental outcomes from standard quantum mechanics in the regimes currently accessible to experiment. Every prediction of quantum optics — interference patterns, complementarity inequalities, delayed-choice correlations, quantum-eraser coincidence counts, interaction-free detection probabilities — is reproduced exactly within the phase-primitive framework. The predictions diverge from standard quantum mechanics only in extreme regimes discussed in the parent theories: Planck-scale coherence effects (where the finite coherence length ℓ_c of the substrate field may become detectable), topological signatures in the cosmic microwave

background (where the global topology of the substrate may imprint observable correlations), and refractive gravity deviations (where the emergent nature of spacetime geometry may produce measurable departures from general relativity at high curvature). These predictions are the subject of ongoing theoretical investigation [30, 31, 32].

Philosophical import. The PNPT demonstrates that the "mystery" of quantum mechanics — at least as it pertains to single-particle interference and measurement — is a mystery of *representation*, not of *nature*. The universe is not paradoxical; our descriptions of it can be, when we import ontological commitments that are inappropriate to the domain. The object-trajectory ontology that works so well for classical mechanics and everyday experience is not a universal ontological framework; when applied to quantum phenomena, it generates paradoxes — not because quantum phenomena are paradoxical, but because object-trajectory ontology is the wrong tool for the job. The phase-primitive framework provides a non-paradoxical alternative that preserves all empirical content while dissolving all conceptual difficulties associated with photon behavior.

Open questions. Several important questions remain beyond the scope of the present paper:

- *Multi-photon entanglement.* Does the PNPT extend to multi-photon entanglement scenarios — GHZ states, W states, cluster states? The framework should generalize naturally (entanglement is substrate continuity in LST, correlated admissibility in Phase Theory), but the formal extension requires development of the multi-excitation sector of the admissible set.
- *Quantum computing and error correction.* How does the phase-primitive framework interface with quantum computation? Quantum gates are unitary operations on the Hilbert space of qubits; in the phase-primitive picture, they correspond to controlled modifications of the admissible set's boundary conditions. Quantum error correction, which protects against decoherence, corresponds to maintaining coherence bounds against environmental constraint incursion. The detailed mapping remains to be worked out.

- *Partial which-path information.* The PNPT establishes that full which-path records and full interference are incompatible ($A_W \cap A_{\text{int}} = \emptyset$). But what happens at intermediate levels of which-path information? The Englert-Greenberger inequality $D^2 + V^2 \leq 1$ allows partial distinguishability and partial visibility. The phase-primitive framework should accommodate this through a continuous family of admissible sets $A(D)$ interpolating between A_{int} ($D = 0$) and A_W ($D = 1$), with the cross-arm coherence terms continuously suppressed as D increases. The formal construction of this interpolation is a natural next step.
- *Gravitational decoherence.* If spacetime is emergent from the phase substrate, gravitational effects may impose additional decoherence constraints. The relationship between the PNPT's admissibility structure and proposals for gravitational decoherence (Penrose, Diósi) merits investigation.

10. Compact Statement

For convenience, we present the paper-ready compact statement of the theorem:

Photon Non-Paradox Theorem.

In phase-primitive frameworks (Phase Theory, LPST, LST), a photon is an equivalence class of admissible phase excitations rather than an object. Interference arises from global phase admissibility; detection is quantized boundary termination; and which-path information corresponds to additional record constraints selecting a decohered admissible sector. Since interference-admissible and record-admissible sectors are disjoint, the standard "photon paradoxes" result only from illicitly importing object-trajectory predicates into a phase-first ontology. Therefore no intrinsic photon paradox exists within these theories.

11. Conclusion

This paper has stated and proved the Photon Non-Paradox Theorem (PNPT), establishing that the standard "photon paradoxes" — wave-particle duality, single-photon self-interference, delayed-choice retrocausality, which-path complementarity, and measurement-induced collapse — are not genuine contradictions within phase-primitive theoretical frameworks. The theorem is grounded in six formal definitions (Section 3), proved constructively through four lemmas (Section 5), and applied to five canonical experiments (Section 7).

The central insight is structural and can be summarized in a single sentence: *photon paradoxes arise from importing object-trajectory predicates into a framework where the photon is not an object but an equivalence class of admissible phase configurations*. Remove the object predicates, and the paradoxes dissolve — not by being answered differently, but by being revealed as ill-formed questions within the phase-primitive ontology.

Four features of the result deserve emphasis:

First, the theorem is constructive. It does not merely assert that paradoxes go away; it shows precisely how each paradox dissolves. Lemma 1 eliminates trajectory identity. Lemma 2 establishes sector exclusivity between which-path and interference sectors. Lemma 3 replaces collapse with boundary termination. Lemma 4 replaces retrocausality with global constraint satisfaction. Each lemma identifies the specific object-predicate whose illicit importation generates the paradox and demonstrates that its removal is both sufficient and non-destructive (no empirical content is lost).

Second, the resolution is unified. All five paradoxes are resolved by a single framework, a single ontological commitment (phase primacy), and a single structural mechanism (the incompatibility of object predicates with phase admissibility). There is no need for separate ad hoc explanations of duality, self-interference, complementarity, delayed choice, and collapse. They are all manifestations of one error: treating the photon as an object.

Third, the resolution is principled. It follows from the ontological commitments of Phase Theory, LPST, and LST — specifically, from the definitions of the phase substrate, admissible configurations, and the consistency functional. No additional axioms, hidden variables, parallel universes, or observer-dependent elements are introduced. The resolution is as parsimonious as the theories from which it derives.

Fourth, the resolution is experimentally grounded. Every prediction of standard quantum optics is reproduced within the phase-primitive framework. The theorem does not sacrifice empirical adequacy for conceptual clarity; it achieves both simultaneously. The double-slit pattern, the Mach-Zehnder output statistics, the delayed-choice correlations, the quantum-eraser coincidence fringes, and the interaction-free detection probabilities are all derived from the admissibility structure without invoking any paradoxical mechanism.

The Photon Non-Paradox Theorem exemplifies a broader program: the derivation of physics from phase topology. In this program, the fundamental entities of physics — particles, fields, forces, spacetime itself — are not primitives but emergent structures arising from the consistency constraints on a phase-bearing substrate. The PNPT shows that this program resolves not only the technical challenges of unification but also the conceptual challenges of interpretation. If the photon is not an object, there is no photon paradox. If spacetime is emergent, there is no retrocausality. If detection is boundary termination, there is no collapse. The paradoxes of quantum mechanics are artifacts of an ontology that the phase-primitive framework transcends.

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